

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26

Year: Second Year

Term: I

Roll No.: 1012411011

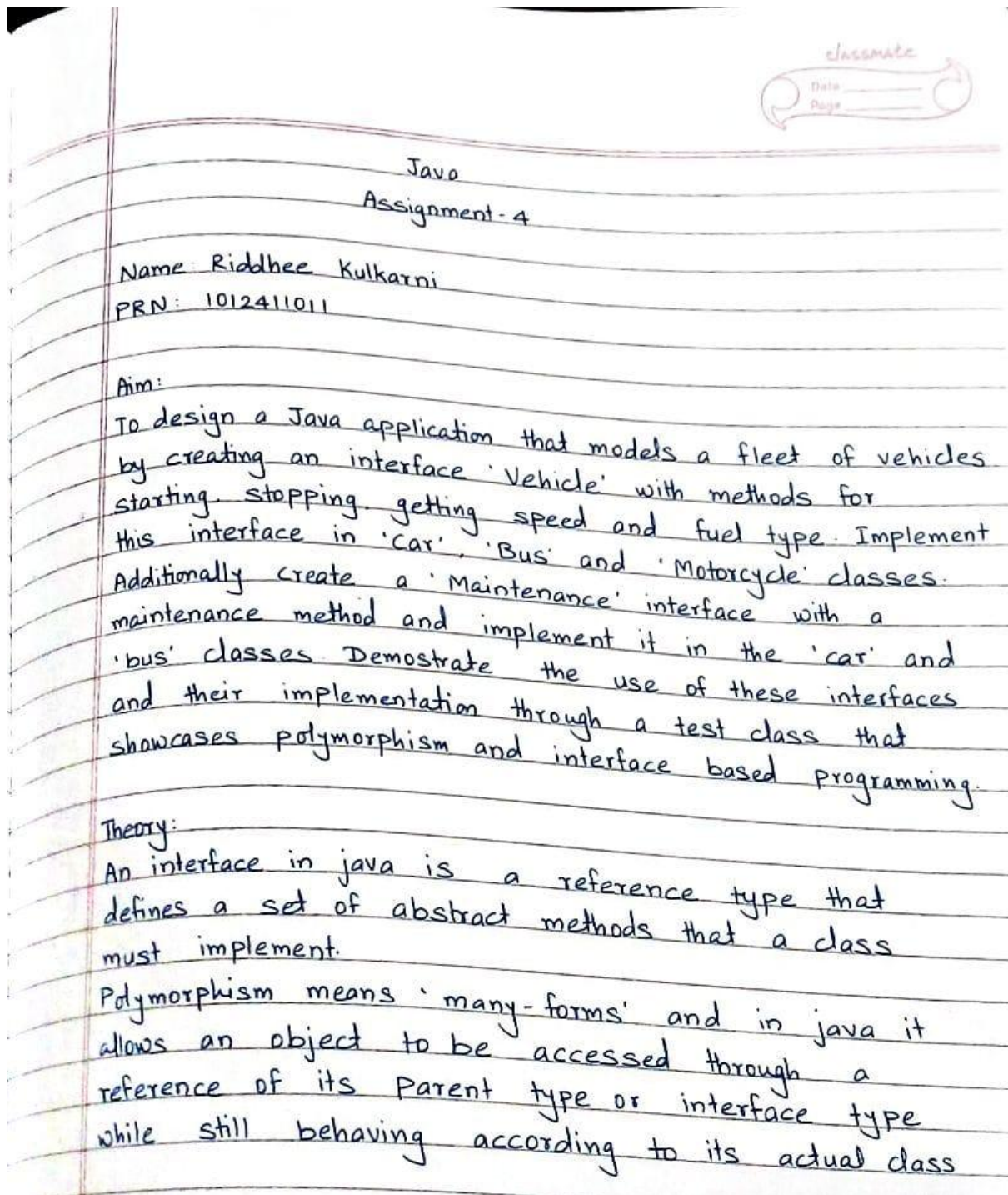
Name: Riddhee Kulkarni

Subject: Java

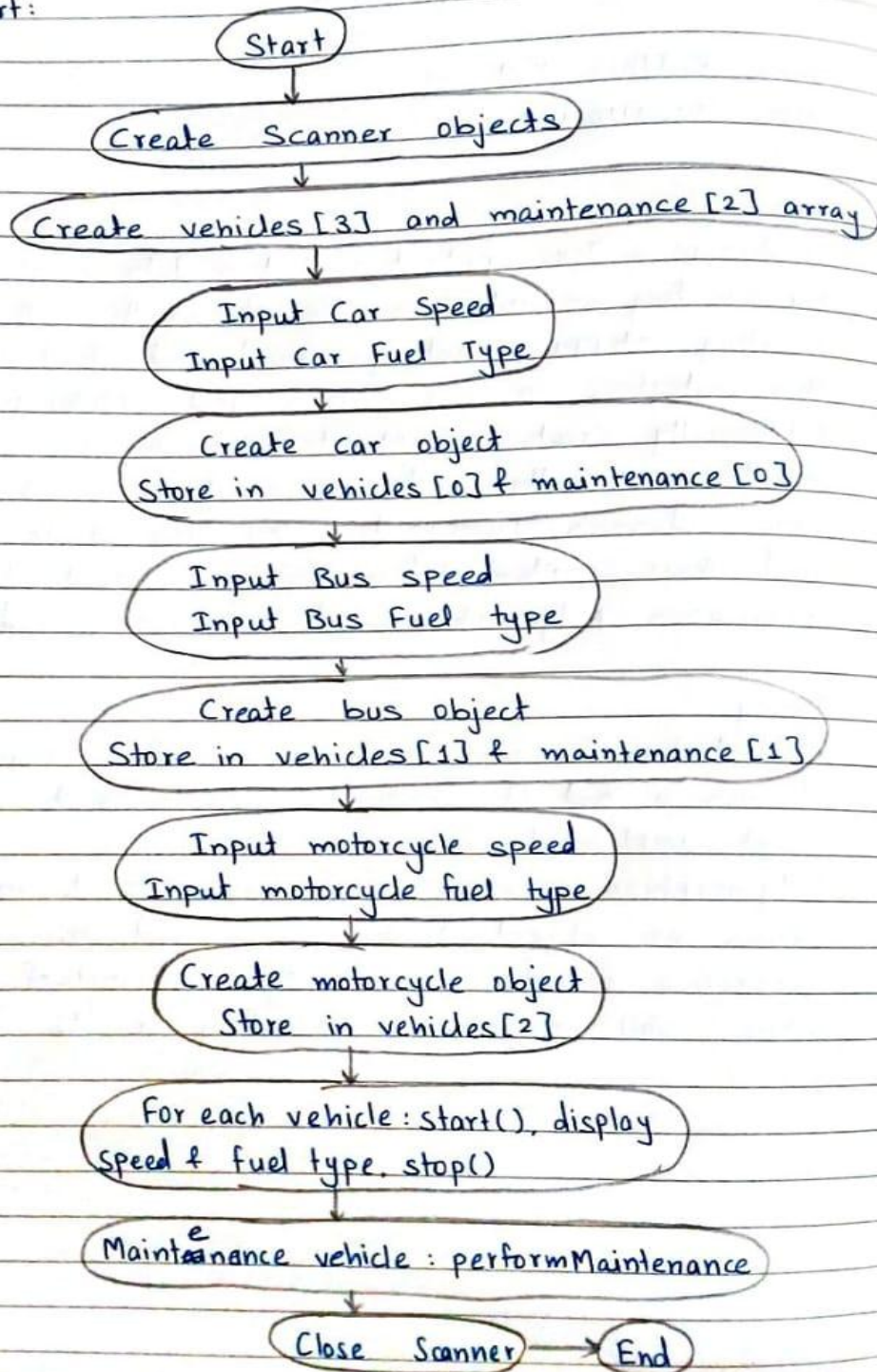
Assignment No.: 4

Title: Vehicle Management System

Date: 13/08/2025



Flowchart:



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```
package Assignment4;
import java.util.Scanner;

interface Vehicle
{
    void start();
    void stop();
    int getSpeed();
    String getFuelType();
}

interface Maintenance
{
    void performMaintenance();
}

class Car implements Vehicle, Maintenance
{
    private int speed;
    private String fuelType;

    Car(int speed, String fuelType)
    {
        this.speed = speed;
        this.fuelType = fuelType;
    }

    public void start()
    {
        System.out.println("Car has started!");
    }

    public void stop()
    {
        System.out.println("Car has been stopped!");
    }

    public int getSpeed()
    {
        return speed;
    }

    public String getFuelType()
    {
        return fuelType;
    }

    public void performMaintenance()
    {
        System.out.println("Car maintenance has been performed!");
    }
}

class Bus implements Vehicle, Maintenance
{
    private int speed;
    private String fuelType;

    Bus(int speed, String fuelType)
    {
        this.speed = speed;
        this.fuelType = fuelType;
    }
}
```

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```
public void start()
{
    System.out.println("Bus has started!");
}

public void stop()
{
    System.out.println("Bus has been stopped!");
}

public int getSpeed()
{
    return speed;
}

public String getFuelType()
{
    return fuelType;
}

public void performMaintenance()
{
    System.out.println("Bus maintenance has been performed!");
}
}

class Motorcycle implements Vehicle
{
    private int speed;
    private String fuelType;

    Motorcycle(int speed, String fuelType)
    {
        this.speed = speed;
        this.fuelType = fuelType;
    }

    public void start()
    {
        System.out.println("Motorcycle has started!");
    }

    public void stop()
    {
        System.out.println("Motorcycle has been stopped!");
    }

    public int getSpeed()
    {
        return speed;
    }

    public String getFuelType()
    {
        return fuelType;
    }
}

public class Interface
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
    }
}
```

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```
Vehicle[] vehicles = new Vehicle[3];
Maintenance[] maintenance = new Maintenance[2];

System.out.println("Enter Car speed :");
int carSpeed = sc.nextInt();
sc.nextLine();
System.out.println("Enter Car fuel type :");
String carFuel = sc.nextLine();
vehicles[0] = new Car(carSpeed, carFuel);
maintenance[0] = (Maintenance) vehicles[0];

System.out.println("Enter Bus speed :");
int busSpeed = sc.nextInt();
sc.nextLine();
System.out.println("Enter Bus fuel type :");
String busFuel = sc.nextLine();
vehicles[1] = new Bus(busSpeed, busFuel);
maintenance[1] = (Maintenance) vehicles[1];

System.out.println("Enter Motorcycle speed :");
int bikeSpeed = sc.nextInt();
sc.nextLine();
System.out.println("Enter Motorcycle fuel type :");
String bikeFuel = sc.nextLine();
vehicles[2] = new Motorcycle(bikeSpeed, bikeFuel);

for (Vehicle v : vehicles)
{
    v.start();
    System.out.println("Speed of the vehicle : " + v.getSpeed() + " km/h");
    System.out.println("Fuel Type of the vehicle : " + v.getFuelType());
    v.stop();
    System.out.println();
}

for (Maintenance m : maintenance)
{
    m.performMaintenance();
}

sc.close();
}
```


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Output:

```
Enter Car speed :  
110  
Enter Car fuel type :  
petrol  
Enter Bus speed :  
100  
Enter Bus fuel type :  
petrol  
Enter Motorcycle speed :  
70  
Enter Motorcycle fuel type :  
petrol  
Car has started!  
Speed of the vehicle : 110 km/h  
Fuel Type of the vehicle : petrol  
Car has been stopped!
```

```
Bus has started!  
Speed of the vehicle : 100 km/h  
Fuel Type of the vehicle : petrol  
Bus has been stopped!
```

```
Motorcycle has started!  
Speed of the vehicle : 70 km/h  
Fuel Type of the vehicle : petrol  
Motorcycle has been stopped!
```

```
Car maintenance has been performed!  
Bus maintenance has been performed!
```

Conclusion:
This application demonstrates how to implement polymorphism and interface based program.